MATH 205

Linear Transformations, Formally

LECTURE 14 · LINEAR TRANSFORMATIONS, FORMALLY

Rotations and shears were only the beginning. The same two rules govern derivatives, integrals, and machines no picture could ever capture.

SAJAL SOHAIL RANA

WELCOME BACK

Where Today Fits

Last time

- Matrices as functions: b = Ax, domain, codomain, range
- A gallery: projection, reflection, dilation, rotation, shear
- Range(T) = Col(A) — our first bridge back to old material

Today

- The formal definition — for ANY vector spaces, not just $\mathbb{R}^n$
- Proving linearity: matrices, derivatives, integrals
- T(0) = 0, and why it's a fast way to spot a fake
- Kernel and range, for transformations with no picture

Course Roadmap

1. Systems of Linear Equations
2. Matrices & Matrix Algebra
3. Determinants
4. Vector Spaces
5. **Linear Transformations**

LECTURE 2 OF 4

The gallery only used $\mathbb{R}^n$. Today the SAME two rules apply to functions, polynomials — anything that's a vector space.

Two short rules, checked four different ways — that's the whole lecture.

7.4 THE FORMAL DEFINITION

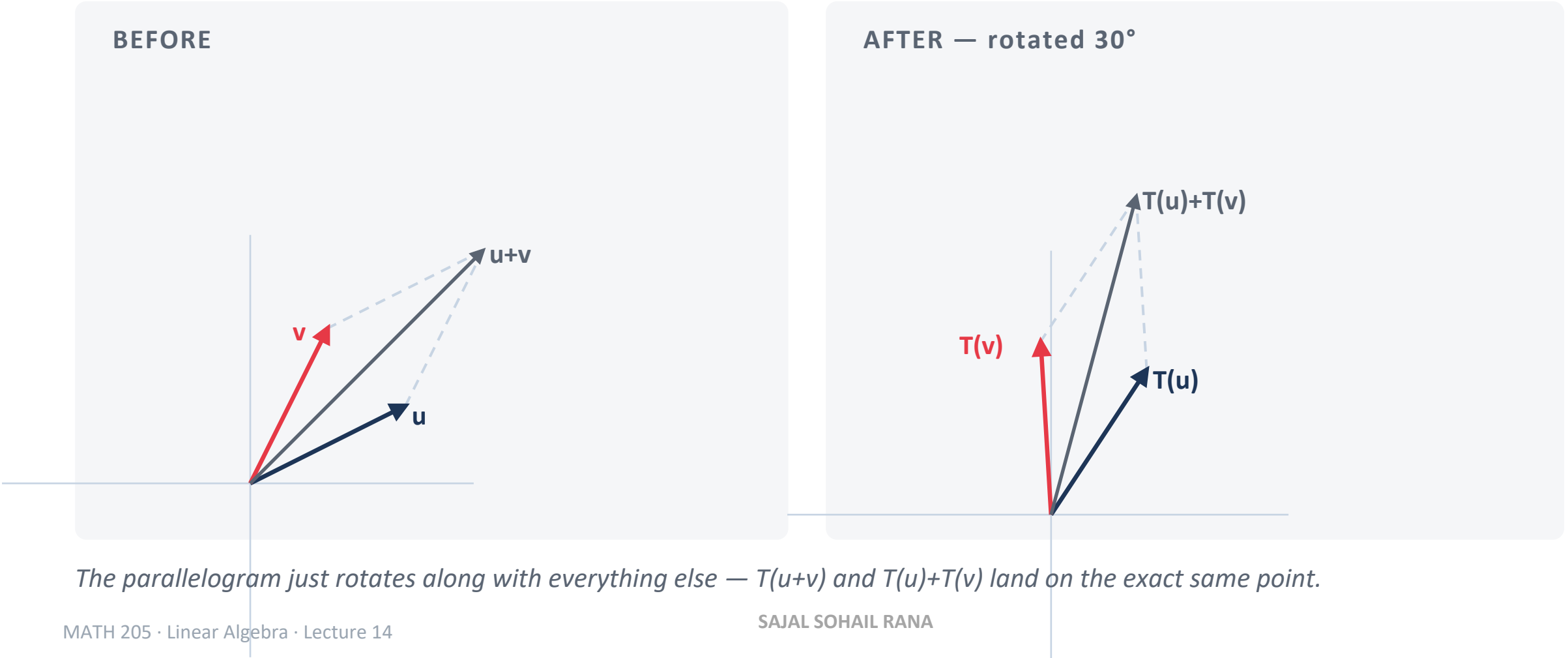
Two Rules, Any Vector Space

Every transformation in last lecture's gallery secretly obeyed the same two rules. Today we write those rules down — and stop needing a picture to use them.

MOTIVATION

What the Gallery had in Common

Take rotation by 30°. Add two vectors first and rotate, or rotate first and add — same answer, either order:



The parallelogram just rotates along with everything else — T(u+v) and T(u)+T(v) land on the exact same point.

7.4 DEFINITION

Linear Transformation

If T : V → W is a function from a vector space V into a vector space W, then T is called a LINEAR TRANSFORMATION if, for all u, v in V and every scalar k:

$$T(\underline{u} + \underline{v}) = T(\underline{u}) + T(\underline{v}) \qquad T(k\underline{u}) = kT(\underline{u})$$

(a) **T(u + v) = T(u) + T(v)** — *ADDITIVITY — sums map to sums*

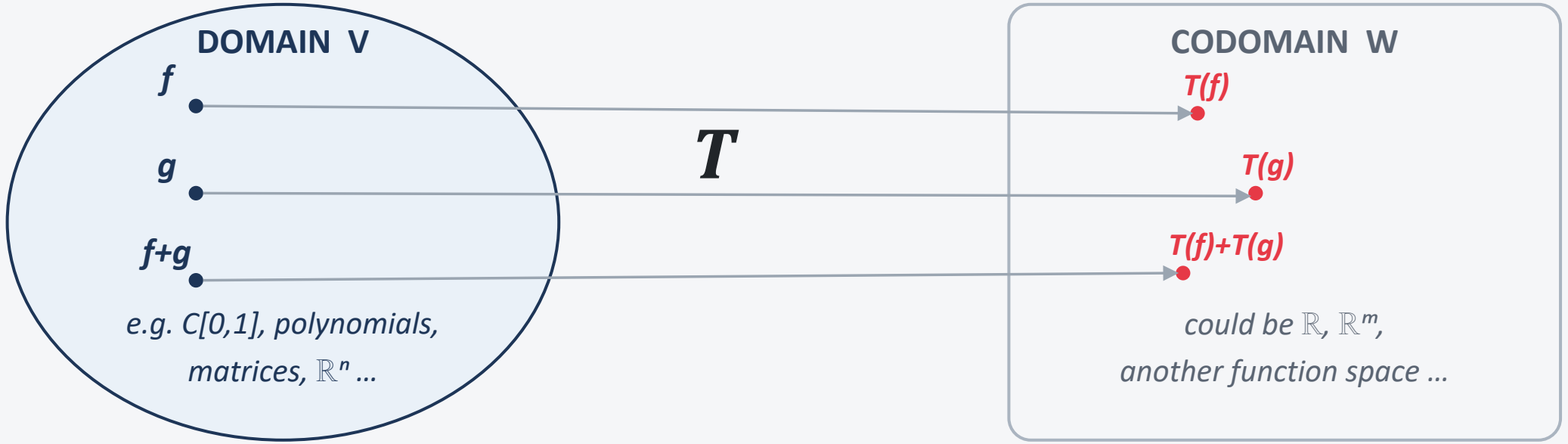
(b) **T(ku) = kT(u)** — *HOMOGENEITY — scaling commutes with T*

V and W can be ANY vector spaces — $\mathbb{R}^n$, polynomials, matrices, even spaces of functions.

PICTURE

Same Diagram, Bigger Spaces

The domain/codomain picture from last lecture still applies — V and W just aren't restricted to $\mathbb{R}^n$ anymore.



7.4 A FIRST CONSEQUENCE

T Always Sends 0 to 0

If T : V → W is a linear transformation, then the zero vector of V always maps to the zero vector of W.

$$T(\underline{0}_V) = \underline{0}_W$$

PROOF: For any v in V, 0 = 0·v (0 the scalar). So:

$$T(\underline{0}) = T(0 \cdot \underline{v}) = 0 \cdot T(\underline{v}) = \underline{0}$$

This gives a fast SANITY CHECK: if a candidate transformation sends 0 to something nonzero, it can't be linear — no further work needed.

7.5 PROVING LINEARITY

Four Transformations, One Recipe

Every proof below follows the same two steps: expand T(u+v), and expand T(ku).

If both collapse cleanly, T is linear.

WORKED EXAMPLE

Matrix Transformations Are Always Linear

$$T(\underline{x}) = A\underline{x}, \qquad T : R^n \to R^m$$

a

$$T(\underline{x}_1 + \underline{x}_2) = A(\underline{x}_1 + \underline{x}_2) = A\underline{x}_1 + A\underline{x}_2 = T(\underline{x}_1) + T(\underline{x}_2)$$

b

$$T(k\underline{x}_1) = A(k\underline{x}_1) = kA\underline{x}_1 = kT(\underline{x}_1)$$

This works for EVERY matrix A — matrix multiplication distributes over addition and commutes with scalars automatically. That's why the entire gallery from last lecture was linear.

WORKED EXAMPLE

Checking a Transformation by Hand

Let f : $\mathbb{R}^2 \to \mathbb{R}^3$ be given by:

$$f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x - y \\ x + y \\ 5x \end{bmatrix}$$

Additivity: let u = (x_1,y_1), v = (x_2,y_2). Then f(u+v) works out to:

$$f(\underline{u} + \underline{v}) = \begin{bmatrix} (x_1+x_2) - (y_1+y_2) \\ (x_1+x_2) + (y_1+y_2) \\ 5(x_1+x_2) \end{bmatrix} = \begin{bmatrix} x_1-y_1 \\ x_1+y_1 \\ 5x_1 \end{bmatrix} + \begin{bmatrix} x_2-y_2 \\ x_2+y_2 \\ 5x_2 \end{bmatrix}$$

= f(u) + f(v) — additivity confirmed.

WORKED EXAMPLE

Finishing the Check

Homogeneity: let u = (x,y) and k be any scalar:

$$f(k\underline{u}) = \begin{bmatrix} kx - ky \\ kx + ky \\ 5kx \end{bmatrix} = k \begin{bmatrix} x - y \\ x + y \\ 5x \end{bmatrix} = kf(\underline{u})$$

Both conditions hold, so f is a LINEAR TRANSFORMATION from $\mathbb{R}^2$ to $\mathbb{R}^3$ — exactly as we'd expect, since f(x,y) = Ax for A = [[1,-1],[1,1],[5,0]].

This is the SAME conclusion the "matrix transformations are always linear" result would have given us instantly — useful when you want to see the mechanics up close.

WORKED EXAMPLE

The Derivative Operator

Let D map a differentiable function to its derivative — no matrix in sight:

$$D(f) = f'(x)$$

a

$$D(f+g) = (f+g)'(x) = f'(x) + g'(x) = D(f) + D(g)$$

(f+g)' = f'+g' — a standard calculus rule.

b

$$D(kf) = (kf)'(x) = kf'(x) = kD(f)$$

Both rules come straight from calculus — so D is linear from W (differentiable functions) to V (all functions).

WORKED EXAMPLE

The Integral Operator

Let J send a continuous function to the (signed) area under its curve:

$$J(f) = \int_0^1 f(x)\,dx, \qquad J : C[0,1] \to R$$

a

$$J(f+g) = \int_0^1 (f+g)(x)\,dx = \int_0^1 f(x)\,dx + \int_0^1 g(x)\,dx = J(f) + J(g)$$

b

$$J(kf) = \int_0^1 kf(x)\,dx = k\int_0^1 f(x)\,dx = kJ(f)$$

Both rules come straight from the basic properties of the definite integral — J is linear from C[0,1] to $\mathbb{R}$.

APPLICATION

Why “Linear” Is Worth Checking

Engineers call additivity + homogeneity the SUPERPOSITION PRINCIPLE — and entire fields are built on systems where it holds.

Circuit analysis	the response to two combined signals is just the sum of the individual responses.
Signal processing	audio filters (like D and J!) can be analyzed one frequency at a time, then added back up.
Structural engineering	the stress from two loads on a beam adds — as long as the material stays in its linear range.
Quantum mechanics	the entire theory rests on states combining via a linear Schrödinger equation.

When a system ISN'T linear (like the robot arm from last lecture), none of these shortcuts apply — which is exactly why checking linearity first saves enormous effort.

7.6 KERNEL

What Gets Sent to Zero?

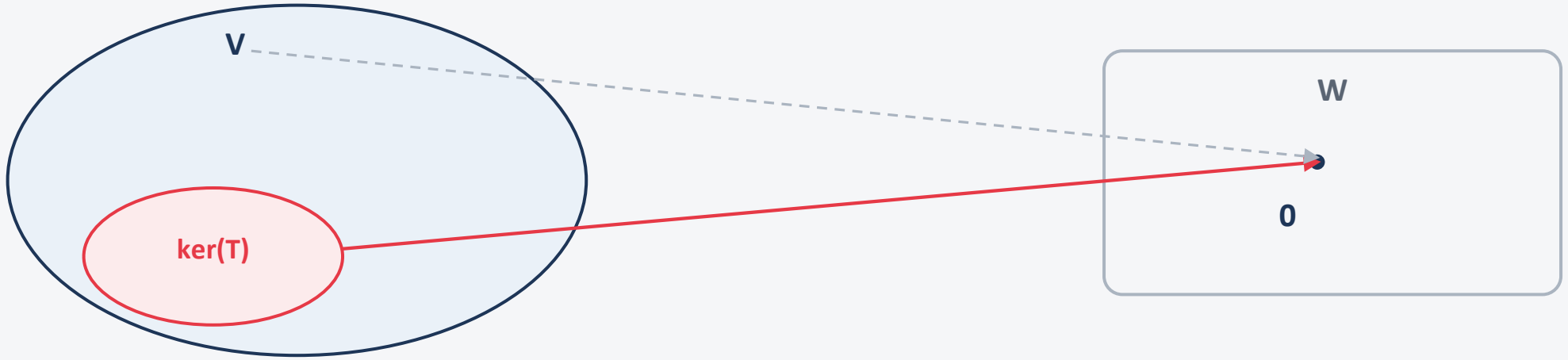
We already know T(0) = 0. The kernel asks the reverse question: which OTHER inputs also collapse to zero?

7.6 DEFINITION

The Kernel of a Linear Transformation

If T : V → W is a linear transformation, the set of vectors in V that map into 0 is called the KERNEL (or nullspace) of T, denoted ker(T).

$$\ker(T) = \{\, \underline{v} \in V \mid T(\underline{v}) = \underline{0} \,\}$$



every vector in ker(T) collapses to the same point: 0

WORKED EXAMPLE

The Kernel of the Derivative

Which functions does D send to the zero function?

$$D(f) = f'(x) = 0 \implies f(x) = k \quad (k \text{ constant})$$

$$\ker(D) = \{\text{all constant functions in } V\}$$

This matches intuition: a function with zero rate of change everywhere is exactly a function that never changes — a constant.

Real-world reading: if all you know is an object's VELOCITY, you can never recover WHERE it started — that starting position is exactly the information the kernel of D throws away.

WORKED EXAMPLE

The Kernel of an Integral

$$J : P_1 \to R, \qquad J(p) = \int_{-1}^{1} p(x)\,dx, \qquad p(x) = a + bx$$

Write p(x) = a + bx and integrate directly:

$$\int_{-1}^{1} (a + bx)\,dx = 2a \implies J(p) = 0 \iff a = 0$$

$$\ker(J) = \{\, p(x) = kx \mid k \in R \,\}$$

The area under p(x)=a+bx on [−1,1] only depends on a (the bx part is an odd function and cancels). So the kernel is every line through the origin.

7.7 RANGE, REVISITED

What Does T Actually Reach?

Last lecture's range = column space idea was special to matrices. Here's the definition that works for any linear transformation at all.

7.7 DEFINITION

Range, for Any Linear Transformation

If T : V → W is a linear transformation, the RANGE of T is the set of all vectors in W that are images under T of at least one vector in V. It is denoted R(T).

$$\text{Range}(T) = \{\, T(\underline{v}) \mid \underline{v} \in V \,\}$$

For a matrix transformation, this is exactly Col(A) — that was never a coincidence, just the special case where V = $\mathbb{R}^n$, W = $\mathbb{R}^m$, and T(x) = Ax. The same definition now covers derivatives, integrals, and everything else.

WORKED EXAMPLE

Finding a Range, Revisited

Recall f(x,y) = (x−y, x+y, 5x) — now that we know it's linear, write it as a matrix:

$$f(\underline{x}) = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 5 & 0 \end{bmatrix} \underline{x}$$

The range is exactly the column space — span the two columns:

$$\text{Range}(f) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Two independent columns in $\mathbb{R}^3$ span a PLANE through the origin — so Range(f) is a 2-dimensional subspace of the 3-dimensional codomain. f is far from onto.

RECAP

Key Takeaways

The Definition

$T(u+v)=T(u)+T(v)$ and $T(ku)=kT(u)$, for ANY vector spaces V and W.

T(0) = 0

an instant test: if a candidate sends 0 elsewhere, it isn't linear.

The Recipe

expand $T(u+v)$ and $T(ku)$, check both collapse to the right form.

Superposition

the real-world name for linearity — circuits, signals, structures, quantum states.

Kernel

$\ker(T) = \{v : T(v) = 0\}$ — the inputs that carry no information through T.

Range

$R(T) = \{T(v) : v \in V\}$ — for matrices, exactly the column space.

NEXT TIME

Looking Ahead

Before Next Lecture

Pick a transformation, find its kernel

Try $T : \mathbb{R}^3 \to \mathbb{R}^2$, T(x,y,z) = (x+y, y+z). Confirm it's linear, then find ker(T) and R(T).

Linking kernel to “onto” and “not one-to-one”

A nonempty kernel is a red flag for “onto” and a green light for “not one-to-one” — next lecture's whole story in one sentence.

Lecture 15

One-to-One, Onto & Inverses

- One-to-one: when ker(T) = {0} — and the pivot-column shortcut.
- Onto: when R(T) fills the whole codomain.
- Inverse linear transformations, and when they exist.

Reference: Anton & Rorres, §8.2–8.3 · ILA §3.2

Questions?

Linear Transformations, Formally · Lecture 14 complete

NEXT: ONE-TO-ONE, ONTO & INVERSES